

Chapter 1: Axiomatics and Rigorous Formalism of TSM: Emergent Time, Chaotic Substrate, and Elastodynamics of the 3-Brane

Before introducing the complete mathematical apparatus describing continuum dynamics and wave phenomena, it is necessary to rigorously define the fundamental parameter with respect to which this dynamics occurs. The Topological 0-Matrix (Substrate) Mechanics (TSM) model completely rejects the concept of empty spacetime and absolute, Newtonian time. The foundation of reality is the Substrate—a dense, stationary, and chaotic ensemble of geometrically confined 0-particles.

1.1. The Fundamental Evolutionary Parameter τ as a Measure of Global, Averaged 0-Particle Activity

In Topological 0-Matrix (Substrate) Mechanics (TSM), we categorically reject both the illusion of absolute Newtonian time and the relativistic "spacetime" devoid of a material carrier. The introduction of a universal reference parameter in this model is not an external imposition of an artificial measure of phenotypic progression; rather, it emerges rigorously and naturally from the vibrating dynamics of the deepest substrate of the Universe.

1.1.1. State Zero as a Stationary Kinetic Chaos

State Zero (a perfectly symmetric configuration devoid of macroscopic deformations and topological structures) is defined as a homogeneous, stationary gas of chaotically colliding 0-particles. We reject the assumption that a crystalline structure is a pre-existing, primary axiom of the system. In State Zero, due to high geometric packing in $\mathbb{R}^4$, all confined 0-particles possess an identical, non-zero intrinsic volume of their oscillation sphere, as well as the same elementary mass and momentum.

Since the ensemble in State Zero is globally stationary and statistically isotropic, the activity of the 0-particles reduces to steady impacts against the boundaries of their own interaction spheres. This leads to the definition of a constant, baseline oscillation frequency of a single 0-particle for an undisturbed system as a statistical average:

$$f_0 \equiv \langle f_{\text{osc}} \rangle_{\text{State Zero}}$$

Differences in all macroscopically observable phenomena (from electro-gravitational interactions to relativistic phenomena) must be rigorously derived from variations in the local collision frequency (f_{lok}). These changes are a direct and exclusively physical consequence of the deformation of the interaction spheres, forcing the 0-particle to undergo more or less frequent reflections due to the shortening or lengthening of its local mean free path.

1.1.2. The Global Evolutionary Parameter τ

Let $N_{\text{total}}(\tau)$ denote the total number of elementary 0-particle impacts against the boundaries of their interaction spheres, summed over all oscillators in the Universe and accumulated from an arbitrarily chosen

reference point. Such mechanics creates an absolute, extensive, and monotonically increasing evolution function of the system:

$$\tau = \frac{N_{\text{total}}}{\mathcal{N}_0 \cdot f_0}$$

where $\mathcal{N}_0$ represents the total number of confined 0-particles in the system. The parameter τ defined in this manner:

- **Emerges from kinetics:** It is an objective mathematical step reflecting the sum of purely mechanical events at the micro-scale.
- **Resolves the synchronization problem:** It relies directly on the geometric summation of statistical collisions isolated within the cages of the oscillators, yielding a smooth, averaged continuity of change at the macro-level.
- **Is measurable exclusively within a thermodynamic abstraction:** No real observer, embedded within the three-dimensional Brane and limited by the intrinsic propagation velocity of waves, has instantaneous access to the total pool of events.

In any arbitrarily short observation interval $d\tau$, an averaged 0-particle residing in a flat, undisturbed region performs $f_0 d\tau$ reflections within its elastic chamber.

1.1.3. Emergence of the Elastic Matrix in the Pre-jamming Regime

The characteristics of a rigid elastic continuum and a macroscopic lattice structure emerge only when the local geometric packing fraction ϕ approaches a critical value ϕ_c . In this pre-jamming state, the freedom of translational motion through the medium vanishes. Only wave-like deformations of stresses (solitons) can propagate through such a highly compressed and mechanically locked Substrate, while the physical nuclei of the oscillators themselves never undergo translation.

The dimensions of the interaction spheres are deeply sub-Planckian, causing the local chaos of fluctuations within the spheres to undergo a rapid process of spatial self-averaging. The governing parameter (the physical hydro-elastic pressure) becomes the local impulse density in evolutionary time, constituting a scalar field:

$$\rho_{\text{imp}}(\mathbf{x}, \tau) = \lim_{V \rightarrow \text{meso}} \frac{1}{V} \sum_{i \in V} \frac{dN_i}{d\tau}$$

As long as a region remains in the undisturbed State Zero, $\rho_{\text{imp}} = \text{const} \cdot f_0$. However, the intrusion of a wave structure or a topological knot into the region drastically constricts the oscillation cage. The invariant momentum of the confined 0-particle causes an immediate spike in the collision rate with the walls of the distorted sphere. The observable value (f_{lok}) deviates upward from the baseline level f_0 . This local collisional jump immediately generates a kinematic pressure differential, serving as the source coupling for all elastodynamic phenomena, wave equations, and gradients upon which the interaction of charges is based.

1.1.4. Measured Time t as a Derivative of Vibration Compaction

Methodological Note (Cross-reference to Chapter 4): The formulation presented here regarding the relationship between the emergent laboratory time t and the evolutionary parameter τ is foundational (kinematic) and pertains to local transformations in a gravitationally undisturbed Substrate. A complete, covariant tensor field formalization—in which the local collision frequency $f_{\text{lok}}(\mathbf{x}, \tau)$ is directly coupled with the macroscopic mechanical stress tensor σ_{ab} , the orthogonal bending of the 3-brane into the x^4 dimension,

and the components of the emergent metric tensor $g_{00}(\mathbf{x})$ —is rigorously derived in Chapter 4 (Sections 4.2–4.5). Within the full TSM field formalism, it will be demonstrated that the classical gravitational potential Φ and the gravitational time dilation conditioned by it are macroscopic descriptions of the hydro-elastic pressure profile of the Substrate.

Under this outlined mechanism, the concept of time is demystified. Time measured in laboratory frameworks has no direct link to the objective vector $N_{\text{total}}(\tau)$. Every local clock (including precise atomic systems) is merely a macroscopic resonator composed of vortex-locked standing wave stresses (solitons), counting its own internal phase equilibrium states.

The instantaneous measure of time passing from the perspective of such a system depends strictly on the confinement state and the degree of deformation of the underlying matrix of cells. The increment of observable time is expressed by the proportion that defines the local dilation factor $dt/d\tau$:

$$\frac{dt}{d\tau} = \frac{f_0}{f_{\text{lok}}(\mathbf{x}, \tau)} \implies dt = \frac{f_0}{f_{\text{lok}}(\mathbf{x}, \tau)} d\tau$$

Where the cells remain undisturbed ($f_{\text{lok}} = f_0$), the observer records a progression of phenomena of the order of $dt = d\tau$. When matter or matrix compaction adjacent to an extreme mass bent toward the 4D axis forces a strong local spatial compression, it results in a shortening of the mean free paths of the vibrating particles and a spike in the collision parameter to $f_{\text{lok}} > f_0$. Consequently, the development of macroscopic processes consumes a smaller pool of event fractions within the system per single global step τ . This constitutes the physical and geometric mechanism of the phenomenon interpreted in classical physics as time dilation.

1.2. Medium and Space: The Geometric Jamming State and the Isotropic 3-Brane

Having defined the evolutionary parameter, we can formally describe space. Fundamental space is a 4-dimensional differentiable manifold $\mathcal{M}^4$. It is not a vacuum, but a pressure vessel completely filled with colliding 0-particles.

1.2.1. The Jamming State

Due to the condition of finiteness of the Universe and extreme packing, in the regime where $\phi \rightarrow \phi_c$, the 0-particles undergo geometric jamming. Each 0-particle becomes confined within its own interaction sphere (Wigner-Seitz cell). This creates a rigorous spatial matrix in 4D, which gains a full stiffness tensor and the capacity to propagate transverse waves. The mechanical state of this medium is described by a symmetric stress tensor σ_{ab} , where the trace of the tensor is denoted as Σ . In the absolute State Zero, prior to the introduction of non-linear defects, the macroscopic shear stress is $\sigma_{ab} = 0$.

1.2.2. The Isotropic 3-Brane

Permanent, high-frequency pressure waves force the self-organization of the Substrate. Our physical Universe is an isotropic, three-dimensional membrane (3-brane) suspended deep within the 4th dimension. Due to statistical averaging of directions at the macroscopic level, the 3-brane behaves as a perfectly continuous, isotropic, non-linear elastic medium.

1.3. Non-linear Kinematics: Emergence of the Metric g_{ab} from Strain

In TSM, the spatial metric g_{ab} is not a primary entity of abstract geometry; it is a physical projection of the strain state of the Substrate. In State Zero, the medium is described by a flat Euclidean metric δ_{ab} . By introducing the physical vector displacement field of the matrix $\phi^a(\mathbf{x})$, the observable (Riemannian) metric emerges rigorously from the non-linear displacement gradients:

$$g_{ab} = \delta_{ab} + \nabla_a \phi_b + \nabla_b \phi_a + \delta_{cd} \nabla_a \phi^c \nabla_b \phi^d$$

The curvature of space and the resulting gravity represent the mechanical stretching of local 0-particle cells, creating a macroscopic stress gradient toward the 4th spatial dimension.

1.4. Substrate Wave Dynamics and Field Equations

The mechanics of the Substrate are governed by a generalized Lagrangian formalism. The Lagrangian density $\mathcal{L}$ of the isotropic 3-brane is defined by its total elastic energy, utilizing the elastic modulus tensor K^{abcd} :

$$\mathcal{L} = \frac{1}{2} K^{abcd} \epsilon_{ab} \epsilon_{cd} + \mathcal{O}(\epsilon^3)$$

The derivative of the Lagrangian with respect to the strain tensor ϵ_{ab} uniquely determines the stress tensor:

$$\sigma^{ab} = \frac{\partial \mathcal{L}}{\partial \epsilon_{ab}} = K^{abcd} \epsilon_{cd}$$

Small field deformations within the 3-brane satisfy a differential wave equation with respect to the evolutionary parameter:

$$\nabla^2 \phi - \frac{1}{c_\perp^2} \frac{\partial^2 \phi}{\partial \tau^2} = \mathcal{S}(\mathbf{x}, \tau)$$

where $c_\perp$ is the propagation velocity of transverse shear waves (identical to the baseline speed of light), and $\mathcal{S}$ is the source function representing the presence of permanent topological deformations. Geometric curvature is a direct reflection of the background stress tensor coupled through the elastic material constants of the Substrate.

1.5. Dual Regime of Elastodynamics: Elastic Waves versus Non-linear Topological Locking

The introduction of a mechanism for instantaneous stress relaxation via geometric spheres of kinetic repulsion generates a fundamental dynamic paradox: how can a stable, permanent material structure (a particle) form and maintain itself in a perfectly elastic medium that, by definition, instantaneously dissipates any excess pressure? If the relaxation time approaches a minimum, a local density gradient should smooth out at wave speed long before the lattice has time to undergo plastic rearrangement.

Resolving this apparent paradox requires a rigorous separation of Substrate dynamics into two complementary regimes, demarcated by a critical packing fraction threshold ϕ_{lock} (the topological locking threshold).

1.5.1. Regime I: Linear Acoustic Relaxation ($\phi < \phi_{\text{lock}}$)

As long as the local compression of the matrix does not exceed the critical stability threshold of the Wigner-Seitz geometric cages, the Substrate behaves as an ideal, linear elastic continuum. In this state:

- Deformations are conservative. The kinetic repulsion spheres undergo deformation (e.g., flattening), but the 0-particles strictly maintain the matrix of their existing neighbors. No alteration of the local matrix topology occurs.
- Excess effective pressure P_{eff} , resulting from an increase in the local collision frequency f_{lok} , is transferred symmetrically outward via a boundary cascade. The medium relaxes energy through the propagation of transverse waves at velocity $c_{\perp}$ and longitudinal waves. The dissipation of disruptive phenomena is instantaneous and complete.

1.5.2. Regime II: Non-linear Topological Locking ($\phi \geq \phi_{\text{lock}}$)

The situation changes drastically when, due to an extreme concentration of energy (e.g., the collision of large-scale pressure waves in 4D space), the system is compressed beyond the plasticity limit ϕ_{lock} .

- The kinetic cages become geometrically distorted. In TSM, the phenomenon of topological modification does not involve the physical translation of 0-particles from one cell to another (the principle of absolute confinement of 0-particles within their own interaction spheres remains inviolate).
- The geometric shape of the cages undergoes drastic deformation. The orientation vectors of the stresses become locally twisted, forcing a permanent rotation of the collisional momentum direction of the confined 0-particles. Instead of transmitting momentum outward, the deformation vectors cross at non-standard angles inside the highly compressed region.
- A kinetic restriction of the freedom of the interaction spheres occurs in the form of a braid: the 0-particles strike the deformed boundaries of their cells so intensely and with such a modified angle of incidence that the stress fields generated in this manner loop into a self-sustaining vortex. Any attempt by the sphere to return to a linear geometric state is immediately blocked by the incoming momentum of an adjacent 0-particle, trapped within the same local rotation of vectors.

In this regime, the Substrate loses its capacity to release spatial stresses through simple acoustic emission. A non-linear braid (soliton) is formed, which gains stability because unraveling this knot would require the system to transition back through a state of extreme compression, thereby creating a massive energy barrier. The paradox is resolved: the medium perfectly propagates elastic waves within its linear limit, but permanently traps energy via the deformation of the shape of the spheres and the rotational twisting of momentum vectors within the topological boundary, giving rise to quantized matter.

1.6. Axiom of Matter: Quantized Topological Knots and the Rigorous Separation of Mass from Charge

Under the influence of extreme non-linear compression, where the local packing fraction of the Substrate exceeds the critical topological locking threshold ($\phi(\mathbf{x}) \geq \phi_{\text{lock}}$), linear fluctuations lose their capacity for wave relaxation. The momentum vectors of the rapid oscillators are forced to geometrically alter their adjacency matrix and loop. Within the 4-dimensional continuum, they form stable standing waves (solitons), which are inherently permanent topological knots identified by macroscopic physics as fermions.

1.6.1. Mathematical Origin of Matter Quantization

The quantized nature of elementary particles is not an ad hoc assumption in TSM, but a strict requirement of the apparatus of algebraic topology. The state of deformation (twisting) of the kinetic cages is described by a vector field of matrix orientation $\mathbf{n}(\mathbf{x})$ (where $|\mathbf{n}| = 1$). For the knot to maintain stability, the field $\mathbf{n}$ far from the core of the braid must smoothly and asymptotically approach an undisturbed, homogeneous background state. This boundary constraint forces the parameter space to close onto a sphere.

Rigorous topological theorems prove that the integral of the density of such spatial twisting can take only integer values. This phenomenon defines an invariant linking number (topological charge) $\mathcal{W} \in \mathbb{Z}$. A knot possessing a charge of $\mathcal{W} = 1$ cannot smoothly radiate away a fraction of its braid, as this would require a global, macroscopic tearing of the continuity of the crystalline matrix network. The threshold ϕ_{lock} acts as the trigger mechanism for confinement, but the rigid rules of spherical geometry enforce the indivisible quantization of the formed solitonic defect.

By virtue of this foundation, TSM rigorously separates the concept of mass from charge:

- **Electric Charge (3D Projection):** It is the directly measured flux of the phase twist vector of the knot, projected onto the plane of our 3-brane. Because a closed twist must always close to a full phase rotation (constituting a geometric multiple of 2π), the charge undergoes absolute quantization to values of $\pm 1e$, regardless of the internal complexity of the knot in the higher dimension.
- **Rest Mass (4D Embedding):** It is defined as the macroscopic equivalent of the elastic energy stored within the strained structure of the knot. It corresponds directly to the depth to which the formed braid orthogonally deflects the 3-brane into the fourth spatial dimension. This ultimately explains mass asymmetry: a complex, multiple topological braid (a proton) forces a much deeper 4D deformation and generates an incomparably stronger background stress gradient than a simple loop (an electron), despite possessing an identical projected charge of $\pm 1e$.

1.6.2. Smooth Motion and Asymptotic Extinction of Discrete Barriers

The propagation of matter through space does not consist of pushing 0-particles through the medium, but rather of the lossless, wave-like transfer of the topological state of entanglement itself through successive interaction spheres.

The spatial size of the knot core (L) drastically exceeds the microscopic size of an individual Wigner-Seitz cell (a). This fundamental asymmetry of scales causes the periodic potential of the crystalline lattice resistance to undergo spatial averaging. This leads to an exponential extinction of the discrete Peierls-Nabarro energy barrier, which dictates the minimum force required to move a defect within a solid:

$$E_{PN} \propto \exp\left(-\pi \frac{L}{a}\right) \approx 0$$

As a result, a topological knot formed in this manner glides within the Substrate in a perfectly smooth fashion, without energy dissipation, thereby rigorously and effortlessly fulfilling Newton's First Law of Motion. The radiation of energy in the form of transverse waves (photons) occurs exclusively during deformations of the knot resulting from collisions or rapid acceleration.

1.7. Derivation of Relativistic Kinematics from Substrate Hydro-elastodynamics

Lorentz transformations and the $E = mc^2$ principle are rigorous consequences of hydrodynamic resistance during the motion of standing waves in a dense continuum. All derivatives and dynamic equations must be strictly projected onto the fundamental evolutionary parameter τ , from which the local time of the observer emerges.

1.7.1. Length Contraction

For a knot moving along the x -axis with a constant evolutionary velocity $v_\tau = \frac{dx}{d\tau}$, we introduce the co-moving coordinate $u = x - v_\tau \tau$. Transforming the homogeneous wave equation requires scaling the x -axis to preserve the isotropic form of the soliton:

$$x' = \frac{u}{\sqrt{1 - \frac{v_\tau^2}{c_\perp^2}}} = \gamma(x - v_\tau \tau)$$

The shape of the knot undergoes mechanical compression by a geometric factor γ under the influence of the elastic resistance of the continuum.

1.7.2. Energy, Momentum, and Force in the Evolutionary Framework

The kinetic and potential strain energies of the knot integrate directly within the space of the Substrate to the values:

$$E(\tau) = \gamma \cdot E_0, \quad m(\tau) = \gamma m_0$$

The vector momentum of the soliton is defined with respect to the global parameter as $p(\tau) = \gamma m_0 v_\tau$. The equation of force acting on the knot within the matrix, after accounting for the local dilation factor of laboratory time $\frac{dt}{d\tau} = \frac{f_0}{f_{\text{lok}}}$, reveals the phenomenon of asymptotic wave resistance when attempting to alter the momentum state:

$$F = \frac{dp}{d\tau} \cdot \left(\frac{dt}{d\tau} \right)^{-1} = m_0 \gamma^3 \cdot \frac{d^2 x}{d\tau^2} \left(\frac{d\tau}{dt} \right)^2$$

As the velocity of the knot approaches the velocity of the shear waves ($v \rightarrow c_\perp$), the factor $\gamma \rightarrow \infty$. The wave resistance of the matrix becomes infinite, preventing the soliton from exceeding the threshold $c_\perp$ on the plane of the 3-brane.

1.8. Variable Wave Propagation (Local $c_\perp$) and Material Constants

The speed of light is not an unalterable universal constant; it follows directly from the non-linear phenomenon of acoustoelasticity of the dense continuum. In a region subjected to mechanical stress (Σ), the local velocity of transverse waves $c_{\perp, \text{lok}}$ is modulated by the non-linear stiffness (Murnaghan constants $\mathcal{A}$):

$$c_{\perp, \text{lok}}^2 = \frac{\mu}{\rho_0} (1 + \mathcal{A}\Sigma)$$

In electrodynamic notation, this translates directly into the effective throughput of the Substrate:

$$c_{\perp, \text{lok}} = \frac{1}{\sqrt{\epsilon_{\text{eff}} \mu_0}}$$

The parameter ϵ_{eff} depends on the local stress gradients induced by magnetic fields or gravitational bending toward the fourth spatial dimension:

$$\epsilon_{\text{eff}} = \epsilon_0 \left(1 + \kappa B^2 + \lambda \frac{\Phi}{c_{\perp}^2} + \dots \right)$$

In the TSM model, hitherto dimensionless physical constants lose their abstract status, becoming mechanothermodynamic parameters and material constants of the 4-dimensional elastic continuum:

- $c_{\perp}$ – the transverse relaxation velocity in the relaxed, pure State Zero.
- κ, ζ, λ – the non-linear elastodynamic coupling constants (magnetic, oscillatory, and gravitational, respectively).
- χ – the topological constant determining the baseline tension and the geometry of knot confinement.
- G_{eff} – the effective gravitational constant, determined by the global degree of relaxation and the background density of the Substrate.

1.9. Chapter 1 Summary

- **New Ontology of State Zero:** State Zero is freed from assumptions regarding a primary crystalline lattice. It is defined as a stationary, homogeneous collisional chaos of sub-Planckian 0-particles. The matrix structure and elasto-mechanical properties emerge as a macroscopic statistical effect exclusively in the regime of high packing density and geometric pre-jamming ($\phi \rightarrow \phi_c$).
- **Emergence of the Evolutionary Parameter τ :** Absolute time is replaced by an objective, monotonically increasing global parameter τ , proportional to the summed number of collisions N_{total} of all oscillators. Laboratory measured time t is a secondary and local parameter, and time dilation arises directly from the alteration of the mean free path of 0-particles within compressed or deformed matrix cells, which shifts the local collision frequency f_{lok} .
- **Dual Dynamic Regime:** The paradox of instantaneous Substrate relaxation is resolved through the introduction of the topological locking threshold ϕ_{lock} . Below this threshold, the continuum behaves linearly and instantaneously dissipates elastic waves ($c_{\perp}$). Above the threshold ($\phi \geq \phi_{\text{lock}}$), a non-linear geometric locking of the interaction spheres occurs, forming permanent, self-sustaining braids and knots (solitons).
- **Rigorous Quantization and Wave Relativism:** The stability and quantization of fermions are derived from topological invariants (linking number $\mathcal{W} \in \mathbb{Z}$). It is demonstrated that relativistic kinematics (Lorentz contraction, increase in elastic mass, the speed-of-light barrier) is not a feature of empty spacetime, but a direct consequence of the hydrodynamic and elastic resistance that the Substrate opposes to moving standing waves.
- **Materialization of Physical Constants:** The speed of light is redefined as the velocity of transverse shear waves ($c_{\perp}$) in the matrix, depending locally on the non-linear stress state of the continuum (acoustoelasticity). All fundamental physical constants (c, e, G, ϵ_0) have acquired the status of measurable material parameters of the 4-dimensional elastic medium.